

Solve:

1. The set of values that makes $P(x)$ true is : $\{2, 4, 6, 8\}$
2. The set of values that makes $Q(x)$ true is : $\{6, 7, 8\}$
3. For every element x in the domain if x is even number then x is greater than 5.
4. False, the implication is False when the hypothesis is true and the conclusion is False.
5. The counterexample is (2) or (4)
 $P(2) \rightarrow Q(2) \Rightarrow$ is False

We have the domain.

$$D = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

And we define the predicates:

$P(x)$: x is an even number

$Q(x)$: $x > 5$

Required:

- 1- Write the set of values that makes $P(x)$ true
- 2- Write the set of values that make $Q(x)$ true
- 3- Convert the statement into a verbal Formulation (English):
 $\forall x \in D, P(x) \rightarrow Q(x)$
- 4- is the statement in (3) true or false? And why?
- 5- Give ~~the~~ a counterexample if the statement is false